

Bala Bhavagna Tummala

📍 Charlotte, NC 📞 +1 (704) 604-3718 [🌐 Bhavagna - LinkedIn](#) ✉ bhavagnatummala@gmail.com

EDUCATION

University of North Carolina at Charlotte
Master of Science in Computer Science

Aug 2023 – May 2025
Charlotte, NC

V R Siddhartha Engineering College
Bachelor of Technology in Computer Science

Jun 2018 – May 2022
Vijayawada, India

TECHNICAL SKILLS

Tech Stack: C, Java, Advanced Python, R, HTML, XML, CSS, Javascript, React, Django, Android Studio, Flutter, Django

Cloud and Databases: Microsoft Azure, MySQL, Oracle, SQL, PostgreSQL, Amazon Web Services (AWS), Firebase

Automation and Version Control: Git, Github

Technologies/Tools: Pega, PowerBI, Object-Oriented Programming, Scrum, Agile, Advanced Excel, Excel Macros, Tableau, Data Structures and Algorithms, Operating Systems, TCP/IP, Natural Language Processing(NLP), Artificial Intelligence(AI), Machine Learning(ML), Deep Learning, Linux, DevOps, Pandas, NumPy, OpenCV, TensorFlow, Keras, ServiceNow

EXPERIENCE

University of North Carolina at Charlotte
Graduate Teaching Assistant

Jan 2025 – Present
Charlotte, NC

- Conducted lab sessions and guided 70+ undergraduate students on database design, query optimization, normalization, and **SQL**.
- Graded assignments and projects, ensuring timely feedback.

Tata Consultancy Services
Systems Engineer

Jun 2022 – Jul 2023
Hyderabad, India

- Developed full-stack applications using **React.js**, enhancing system performance through efficient UI components, seamless backend integration.
- Designed, implemented, and maintained **RESTful APIs** to facilitate seamless communication between the **React frontend** and backend services, reducing API response times by 40% and improving data exchange efficiency.
- Successfully led the transformation of **Pega**-based solutions for high-profile financial banking clients, and ensured alignment with business objectives, resulting in a 25% increase in operational efficiency.
- Conducted a thorough evaluation of system requirements, leading to a 30% reduction in project delivery time, translated them into clear and actionable technical specifications, enhancing project clarity and reducing requirement-related changes by 40%.
- Spearheaded continuous operational evaluation of cloud-based applications with targeted **Azure** Monitoring tools, reducing average response times by an impressive 20 seconds per incident.
- Efficiently resolved over 95% of customer incident tickets within SLA using **ServiceNow**, elevating customer satisfaction scores accordingly.

Indian Servers

Sep 2020 – May 2021
Vijayawada, India

ML and Power BI Intern

- Successfully generated and deployed predictive analytics models using **LSTM**, increasing forecast accuracy by 20%, optimizing model performance by 50%, and reducing model size by 40%, significantly enhancing system efficiency and decision-making.
- Analyzed and processed large datasets using **Pandas**, **NumPy**, and **SQL**, ensuring data quality across platforms for more accurate decision-making.
- Automated **data extraction**, transformation, and loading **ETL** processes to ensure clean and accurate data pipelines, improving reporting accuracy.
- Utilized **Power BI** and **Advanced Excel** for interactive reports, enhancing data-driven decision-making and reporting efficiency by 30%.

PROJECT

Diabetic Retinopathy Detection | Python, Machine Learning, Tensorflow, Convolutional Neural Networks

- Proposed a deep learning model based on EfficientNet, achieving 95% accuracy in Diabetic Retinopathy detection. This project involved optimizing EfficientNet for retinal image analysis, significantly boosting early-stage disease identification. This advancement marks a significant contribution to AI in medical diagnostics.

Voice-Based Email for Blind | Python, Artificial Intelligence, Natural Language Processing, Django

- Engineered a voice-based email system using AI, **Python**, and **Django** to make email services accessible for visually challenged individuals.
- Implemented advanced **NLP** algorithms combined with text-to-speech features to create a hands-free experience; saved users an average of 30 minutes per day by streamlining email interactions without visual input.

Kisaan Sahay | Java, Flutter, Firebase, Android Studio

- Developed a machinery-sharing application using **Flutter** and **Firebase**, creating a platform that connected over 500 farmers with necessary agricultural equipment, resulting in improved accessibility and reduced downtime for essential tasks.
- Created a user-friendly interface and real-time database management system for seamless navigation and secure transactions. This project benefited over 500 farmers by reducing operational costs and improving equipment accessibility.

ACHIEVEMENTS

Publications: Voice-Based Email for Blind using AI and Python, Kisan Sahay app.

Awards: 2nd Place – Technical Model Exhibition, Tech Expo, VRSEC

Certifications: Pega Certified Senior System Architect(PCSSA), Azure Fundamentals, Cisco CCNA, Netacad: Cybersecurity Essentials, Nptel: Programming in R, AWS Academy Graduate - AWS Data Engineering, Big Data 101